الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 6 · LESSONS 2-2 · 2-3 · 2-4

Weekly Quiz Preparation

Grade 10 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3 4 5

5	10	40	15
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Friday 9 October 2026**

One page per objective · every solution written out step by step

Quiz map

5 objectives = 5 recipes · 5 أهداف = 5 وصفات

1 **Diagonals of a parallelogram** الأقطار
They bisect each other: equal halves → equation

2 **Is it a parallelogram?** الاختبارات
Pick ONE test (2.9 · 2.10 · 2.11 · 2.12) that fits the data

3 **Parallelogram with coordinates** المستوى الإحداثي
Slope · distance · midpoint → compare; D = A + C − B

4 **Rectangle measures** المستطيل
Four 90° angles · diagonals congruent and bisect · Pythagoras

5 **Is it a rectangle?** إثبات المستطيل
Parallelogram + a right angle (slopes × = −1) or congruent diagonals

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Diagonals of parallelograms

أقطار متوازيات الأضلاع

Lesson 2-2 · Part 2 · pp. 67–74
MAT.3.09.01.011

Theorem: the diagonals of a parallelogram bisect each other — so each diagonal is cut into two equal halves.

The diagonals of a parallelogram bisect each other: each diagonal is cut into two equal parts.

قطرا متوازي الأضلاع ينصّف كلٌّ منهما الآخر.

Recipe الوصفة

1. **Name the point E where the diagonals meet**
   سمِّ نقطة التقاطع E
2. **Equal halves: AE = EC and BE = ED**
   النصفان متساويان
3. **Write the equation from the lengths given, then solve**
   اكتب المعادلة وحلّها
4. **Angles: use vertical angles, a linear pair, alternate interior angles**
   الزوايا: متقابلة بالرأس، مستقيمة، تبادلية

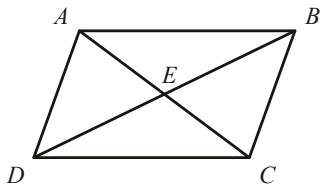
Watch out

- Equal halves, not equal diagonals (that is a rectangle).
- Check by putting the value back into both halves.

Tool. vertical angles are equal · a linear pair adds to 180° · alternate interior angles are equal

Worked example Authored (shape of Ex 9)

In parallelogram $ABCD$ the diagonals meet at E. If $AE = 3x + 1$ and $EC = 5x - 7$, find x and the length of $\overline{AC}$.



1. the diagonals bisect each other
   $AE = EC$
2. write the equation
   $3x + 1 = 5x - 7$
3. subtract 3x, add 7, divide by 2
   $8 = 2x \Rightarrow x = 4$
4. AC is the whole diagonal: two halves
   $AE = 3(4) + 1 = 13, \qquad AC = 2(13) = 26$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 9) In parallelogram $ABCD$ the diagonals meet at E. If $AE = 2z + 5$ and $EC = z + 9$, find z and AC.
2. Authored (shape of Ex 21) In parallelogram $ABCD$ the diagonals meet at F, and $m\angle AFB = 117°$. Find $m\angle BFC$.
3. Authored (shape of Ex 22) In parallelogram $ABCD$, $m\angle ABD = 35°$. Find $m\angle BDC$ and give the reason.

2 Tests for parallelograms

اختبارات متوازي الأضلاع

Lesson 2-3 · Part 1 · pp. 75–82
MAT.3.09.01.012

To prove a quadrilateral is a parallelogram you only need ONE of the tests — choose the one that fits the data.

A quadrilateral is a parallelogram if both pairs of opposite sides (or angles) are congruent, or the diagonals bisect each other.

يكون الشكل متوازي أضلاع إذا تطابق كل ضلعين متقابلين أو كل زاويتين متقابلتين أو نصّف القطران بعضهما.

Recipe الوصفة

1. **Pick the test that matches the given data**
   اختر الاختبار المناسب
2. **Opposite sides ≅ (2.9) · opposite angles ≅ (2.10)**
   أضلاع أو زوايا متقابلة
3. **Diagonals bisect each other (2.11) · one pair ≅ and ∥ (2.12)**
   القطران ينصّف أحدهما الآخر
4. **Set the expressions equal, solve, state the theorem**
   ساوِ وحلّ واذكر النظرية

Watch out

- One pair of opposite sides congruent is NOT enough (an isosceles trapezoid has it).
- Pair the opposite sides: AB with CD, BC with DA.

Tool. Theorems 2.9 · 2.10 · 2.11 · 2.12 — each one names a way to show “parallelogram”

Worked example Authored (shape of Ex 8)

Find x and y so that $ABCD$ is a parallelogram: $AB = 5x - 4$, $CD = 3x + 6$, $BC = y + 8$, $DA = 2y + 3$.

1. opposite sides congruent (Theorem 2.9)
   $AB = CD$: $\quad 5x - 4 = 3x + 6$
2. subtract 3x, add 4, divide by 2
   $2x = 10 \Rightarrow x = 5$
3. the other pair of opposite sides
   $BC = DA$: $\quad y + 8 = 2y + 3$
4. subtract y, subtract 3
   $y = 5$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 7) In quadrilateral $ABCD$: $m\angle A = (2x + 10)°$, $m\angle C = (3x - 20)°$, $m\angle B = (y + 40)°$, $m\angle D = (2y - 10)°$. Find x and y so that it is a parallelogram.
2. Authored (shape of Ex 1-6) In quadrilateral $ABCD$, $\overline{AB} \parallel \overline{DC}$ and $AB = DC$. Is $ABCD$ a parallelogram? Give the theorem.
3. Authored (shape of Ex 1-6) A quadrilateral has diagonals of 8 cm and 6 cm that meet at their midpoints. Is it a parallelogram? Why?

Three formulas do the work: Slope (parallel), Distance (congruent), Midpoint (bisecting diagonals).

Use the Slope, Distance or Midpoint Formula to show a quadrilateral is a parallelogram, or to find the fourth vertex.

استخدم صيغة الميل أو المسافة أو المنتصف لإثبات متوازي الأضلاع أو لإيجاد الرأس الرابع.

Recipe الوصفة

1. **Choose a test: slopes · distances · midpoints** — اختر الصيغة
2. **Compute it for both pairs of opposite sides, or for both diagonals** — احسبها للأزواج المتقابلة
3. **Compare: equal slopes, equal lengths, same midpoint** — قارن
4. **State the conclusion with the theorem** — اكتب النتيجة

Watch out

- Midpoint: add the x's and the y's, then halve.
- Missing vertex: D = A + C − B (the diagonal sum).

Tool. slope $m = (y_2 - y_1)/(x_2 - x_1)$ · midpoint $((x_1+x_2)/2, (y_1+y_2)/2)$ · distance $\sqrt{((\Delta x)^2 + (\Delta y)^2)}$

Worked example Authored (shape of Ex 14)

Use the Midpoint Formula to determine whether $A(0,0)$, $B(4,1)$, $C(5,4)$, $D(1,3)$ form a parallelogram.

1. midpoint of diagonal AC

$$\text{midpoint of } \overline{AC} = \left(\frac{0+5}{2}, \frac{0+4}{2}\right) = \left(\frac{5}{2}, 2\right)$$

2. midpoint of diagonal BD

$$\text{midpoint of } \overline{BD} = \left(\frac{4+1}{2}, \frac{1+3}{2}\right) = \left(\frac{5}{2}, 2\right)$$

3. same midpoint → they bisect each other (Theorem 2.11)

the diagonals bisect each other $\Rightarrow$ $ABCD$ is a parallelogram

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 12) Use the Slope Formula to determine whether $P(1,1)$, $Q(5,2)$, $R(6,5)$, $S(2,4)$ form a parallelogram.
2. Authored (shape of Ex 20) $ABCD$ is a parallelogram with $A(2,1)$, $B(6,2)$, $C(7,6)$. Find D.
3. Authored (shape of Ex 14) Use the Midpoint Formula to determine whether $W(0,0)$, $X(4,1)$, $Y(6,5)$, $Z(1,3)$ form a parallelogram.

4 Properties of rectangles

خصائص المستطيل

Lesson 2-4 · Part 1 · pp. 83–89
MAT.3.09.01.014

A rectangle is a parallelogram with four right angles — so its diagonals are congruent AND bisect each other.

A rectangle has four right angles; its diagonals are congruent and bisect each other.

للمستطيل أربع زوايا قائمة، وقطراه متطابقان وينصّف كلٌّ منهما الآخر.

Recipe الوصفة

1. **Opposite sides congruent; all angles 90°**
   أضلاع متقابلة متطابقة وزوايا قائمة
2. **Diagonals congruent: AC = BD**
   القطران متطابقان
3. **Diagonals bisect each other: AE = EC = BE = ED**
   ينصّف كلٌّ منهما الآخر
4. **A diagonal from two sides: Pythagorean Theorem**
   القطر من الضلعين: فيثاغورس

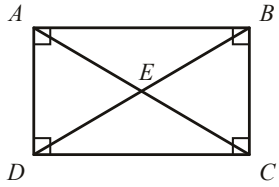
Watch out

- Congruent diagonals AND bisecting diagonals (a parallelogram only bisects).
- A diagonal makes a right triangle: legs are the sides.

Tool. Theorem 2.13: the diagonals of a rectangle are congruent · $a^2 + b^2 = c^2$

Worked example Authored (shape of Ex 29-30)

Rectangle $ABCD$ has $AB = 8$ and $BC = 6$, and its diagonals meet at E. Find AC and BE.



1. triangle ABC has a right angle at B: Pythagorean Theorem
   $AC^2 = AB^2 + BC^2 = 8^2 + 6^2 = 100$
2. take the positive square root
   $AC = 10$
3. the diagonals are congruent
   $BD = AC = 10$
4. the diagonals bisect each other: E is the midpoint
   $BE = \frac{1}{2}BD = 5$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 12) The diagonals of rectangle $EFGH$ meet at K. If $EK = 3x + 2$ and $KG = 5x - 6$, find x and EG.
2. Authored (shape of Ex 2) In rectangle $ABCD$, $m\angle CAB = 35°$. Find $m\angle ACB$.
3. Authored (shape of Ex 11) The diagonals of rectangle $ABCD$ are $AC = 2x + 4$ and $BD = 3x - 3$. Find x and AC.

5 Proving a rectangle

إثبات أن الشكل مستطيل

First a parallelogram, then ONE more fact: a right angle, or congruent diagonals.

A parallelogram is a rectangle if it has a right angle or congruent diagonals (Theorem 2.14).

متوازي الأضلاع مستطيل إذا كانت فيه زاوية قائمة أو كان قطراه متطابقين.

Recipe الوصفة

1. **Show it is a parallelogram (slopes or midpoints)**
   أثبت أنه متوازي أضلاع
2. **Right angle: the slopes of two consecutive sides multiply to −1**
   زاوية قائمة: ناتج ضرب الميلين = −1
3. **Or diagonals: the Distance Formula gives AC = BD**
   أو القطران: صيغة المسافة
4. **State: a parallelogram with a right angle is a rectangle**
   اكتب النتيجة

Watch out

- Slopes that multiply to −1 give a RIGHT ANGLE, not parallel sides.
- No right angle (product ≠ −1) means “not a rectangle”.

Worked example Authored (shape of Ex 17-22)

Use the Slope Formula to determine whether $J(-2,0)$, $K(2,-2)$, $L(5,4)$, $M(1,6)$ are the vertices of a rectangle.

1. slopes of one pair of opposite sides

$$JK:\ \frac{-2-0}{2+2}=-\frac{1}{2},\qquad ML:\ \frac{4-6}{5-1}=-\frac{1}{2}$$

2. slopes of the other pair: parallelogram

$$KL:\ \frac{4+2}{5-2}=2,\qquad JM:\ \frac{6-0}{1+2}=2$$

3. consecutive sides: slopes multiply to −1

$$\left(-\frac{1}{2}\right)(2)=-1 \Rightarrow \overline{JK}\perp\overline{KL}$$

4. conclusion

a parallelogram with a right angle: a rectangle

Your turn · solve in your notebook · solutions at the back

1. **Authored (shape of Ex 19)** Use the Slope Formula to determine whether $A(0,0)$, $B(5,1)$, $C(7,4)$, $D(2,3)$ are the vertices of a rectangle.
2. **Authored (shape of Ex 24)** A parallelogram has diagonals of 12 cm and 12 cm. Is it a rectangle? Give the theorem.

3. **Authored (shape of Ex 21)** Use the Distance Formula to determine whether $P(-1,-1)$, $Q(3,1)$, $R(2,3)$, $S(-2,1)$ are the vertices of a rectangle.

Objective 1 · Diagonals of parallelograms أقطار متوازيات الأضلاع

1.1 Authored (shape of Ex 9)

In parallelogram $ABCD$ the diagonals meet at E. If $AE = 2z + 5$ and $EC = z + 9$, find z and AC.

1. halves are equal: write the equation
   $AE = EC \Rightarrow 2z + 5 = z + 9$
2. subtract z, subtract 5
   $z = 4, \qquad AE = 2(4) + 5 = 13$
3. $AC = AE + EC = 2 \cdot 13$
   $AC = 2(13) = 26$

1.2 Authored (shape of Ex 21)

In parallelogram $ABCD$ the diagonals meet at F, and $m\angle AFB = 117°$. Find $m\angle BFC$.

1. they lie along the diagonal AC
   $\angle AFB$ and $\angle BFC$ are a linear pair
2. a linear pair adds to $180°$
   $m\angle BFC = 180 - 117 = 63°$

1.3 Authored (shape of Ex 22)

In parallelogram $ABCD$, $m\angle ABD = 35°$. Find $m\angle BDC$ and give the reason.

1. opposite sides of a parallelogram are parallel
   $\overline{AB} \parallel \overline{DC}$
2. alternate interior angles are equal (transversal BD)
   $m\angle BDC = m\angle ABD = 35°$

Objective 2 · Tests for parallelograms اختبارات متوازي الأضلاع

2.1 Authored (shape of Ex 7)

In quadrilateral $ABCD$: $m\angle A = (2x + 10)°$, $m\angle C = (3x - 20)°$, $m\angle B = (y + 40)°$, $m\angle D = (2y - 10)°$. Find x and y so that it is a parallelogram.

1. opposite angles congruent (Theorem 2.10)
   $m\angle A = m\angle C: \quad 2x + 10 = 3x - 20$
2. subtract $2x$, add 20
   $x = 30$
3. the other pair of opposite angles
   $m\angle B = m\angle D: \quad y + 40 = 2y - 10$
4. subtract y, add 10
   $y = 50$

2.2 Authored (shape of Ex 1-6)

In quadrilateral $ABCD$, $\overline{AB} \parallel \overline{DC}$ and $AB = DC$. Is $ABCD$ a parallelogram? Give the theorem.

1. one pair of opposite sides is parallel AND congruent
   $\overline{AB} \parallel \overline{DC}$ and $AB = DC$
2. Theorem 2.12
   $ABCD$ is a parallelogram

2.3 Authored (shape of Ex 1-6)

A quadrilateral has diagonals of 8 cm and 6 cm that meet at their midpoints. Is it a parallelogram? Why?

1. they meet at their midpoints (the lengths 8 and 6 are not needed)
   the diagonals bisect each other
2. Theorem 2.11
   so it is a parallelogram

Objective 3 · Parallelograms in the coordinate plane — متوازيات الأضلاع في المستوى الإحداثي

3.1 Authored (shape of Ex 12)

Use the Slope Formula to determine whether $P(1,1)$, $Q(5,2)$, $R(6,5)$, $S(2,4)$ form a parallelogram.

1. slopes of one pair of opposite sides
   $PQ: \dfrac{2-1}{5-1} = \dfrac{1}{4}, \qquad SR: \dfrac{5-4}{6-2} = \dfrac{1}{4}$
2. slopes of the other pair
   $QR: \dfrac{5-2}{6-5} = 3, \qquad PS: \dfrac{4-1}{2-1} = 3$
3. definition of a parallelogram
   both pairs of opposite sides are parallel: yes

3.2 Authored (shape of Ex 20)

$ABCD$ is a parallelogram with $A(2,1)$, $B(6,2)$, $C(7,6)$. Find D.

1. ABCD: B and D are opposite, so D = A + C − B
   $D = A + C - B$
2. add the coordinates, then subtract
   $D = (2+7-6,\ 1+6-2) = (3,\ 5)$

3.3 Authored (shape of Ex 14)

Use the Midpoint Formula to determine whether $W(0,0)$, $X(4,1)$, $Y(6,5)$, $Z(1,3)$ form a parallelogram.

1. midpoint of diagonal WY

$$\text{midpoint of } \overline{WY} = \left(3, \frac{5}{2}\right)$$

2. midpoint of diagonal XZ

$$\text{midpoint of } \overline{XZ} = \left(\frac{5}{2}, 2\right)$$

3. the diagonals do not bisect each other

different midpoints: not a parallelogram

Objective 4 · Properties of rectangles خصائص المستطيل

4.1 Authored (shape of Ex 12)

The diagonals of rectangle $EFGH$ meet at K. If $EK = 3x + 2$ and $KG = 5x - 6$, find x and EG.

1. the diagonals bisect each other

$EK = KG: \quad 3x + 2 = 5x - 6$

2. subtract 3x, add 6, divide by 2

$x = 4, \qquad EK = 3(4) + 2 = 14$

3. EG is the whole diagonal

$EG = 2(14) = 28$

4.2 Authored (shape of Ex 2)

In rectangle $ABCD$, $m\angle CAB = 35°$. Find $m\angle ACB$.

1. a rectangle has four right angles

$m\angle ABC = 90°$

2. the angles of triangle ABC add to 180°

$m\angle ACB = 180 - 90 - 35 = 55°$

4.3 Authored (shape of Ex 11)

The diagonals of rectangle $ABCD$ are $AC = 2x + 4$ and $BD = 3x - 3$. Find x and AC.

1. the diagonals of a rectangle are congruent

$AC = BD: \quad 2x + 4 = 3x - 3$

2. subtract 2x, add 3

$x = 7, \qquad AC = 2(7) + 4 = 18$

Objective 5 · Proving a rectangle — إثبات أن الشكل مستطيل

5.1 Authored (shape of Ex 19)

Use the Slope Formula to determine whether $A(0,0)$, $B(5,1)$, $C(7,4)$, $D(2,3)$ are the vertices of a rectangle.

1. opposite sides have equal slopes: a parallelogram

$AB: \frac{1}{5}$, $DC: \frac{4-3}{7-2}=\frac{1}{5}$; $BC: \frac{4-1}{7-5}=\frac{3}{2}$, $AD: \frac{3}{2}$

2. consecutive sides: the product is 3/10

$\frac{1}{5}\cdot\frac{3}{2}=\frac{3}{10}$ $(\text{not } -1)$

3. not perpendicular, so not a rectangle

no right angle: not a rectangle

5.2 Authored (shape of Ex 24)

A parallelogram has diagonals of 12 cm and 12 cm. Is it a rectangle? Give the theorem.

1. a parallelogram with congruent diagonals

the diagonals are congruent (12 cm)

2. Theorem 2.14

so it is a rectangle

5.3 Authored (shape of Ex 21)

Use the Distance Formula to determine whether $P(-1,-1)$, $Q(3,1)$, $R(2,3)$, $S(-2,1)$ are the vertices of a rectangle.

1. opposite sides are congruent: a parallelogram

$PQ^2=20$, $RS^2=20$; $QR^2=5$, $SP^2=5$

2. the diagonals PR and QS

$PR^2=3^2+4^2=25$, $QS^2=(-5)^2+0^2=25$

3. congruent diagonals: Theorem 2.14

opposite sides congruent, diagonals congruent: a rectangle

Ready?

Tick each line · ضع علامة

- [ ] **I can use the diagonals of a parallelogram (halves are equal)**
  أستطيع استخدام أقطار متوازي الأضلاع
- [ ] **I can choose the test that proves a parallelogram**
  أستطيع اختيار اختبار متوازي الأضلاع
- [ ] **I can use slope, distance and midpoint in the coordinate plane**
  أستطيع استخدام الصيغ في المستوى الإحداثي
- [ ] **I can find measures in a rectangle**
  أستطيع إيجاد القياسات في المستطيل
- [ ] **I can prove that a parallelogram is a rectangle**
  أستطيع إثبات أن متوازي الأضلاع مستطيل

On the day

Draw the figure and mark what is given before you write an equation. Write the first line at once: the first step already earns a mark.

Every question

Name the theorem or formula you use. In the coordinate plane show the formula and the numbers you put into it.

One mark for every written step

لكل خطوة تكتبها علامة